

Resource-Normalized Scaling Laws for Byzantine Consensus: Chance-Constrained Validator-Set Sizing from Commodity-CI Measurements

Peniak B. O., Liubinskiy B. B.

¹*Lviv Polytechnic National University
12 S. Bandera str., 79013, Lviv, Ukraine*

²*Institute of Applied Mathematics and Fundamental Sciences*

(Received 29 July 2026)

Sizing a permissioned BFT validator set for e-governance trades consensus throughput, which falls with n , against fault-detection power, which rises with it. We propose resource-normalized measurement with a fixed per-validator CPU quota; show that, in the unsaturated bi-factor model with zero-mean noise, the single-factor OLS exponent tends to $\gamma\lambda - \beta$ (an omitted-variable bias); and formulate chance-constrained sizing with feasible set $[n_{\min}, n_{\max}]$. On four-core public CI ($n \leq 16$) the exponent is stable at a fixed quota but quota-dependent; the measured detector is near chance (AUC = 0.506), so the detection constraint is relaxed (not enforced) and throughput alone sets the window. Coefficients are deployment-specific.

Keywords: *Byzantine fault tolerance, consensus, scaling law, capacity planning, chance-constrained optimization, e-governance*

2000 MSC: 68M14, 68W20, 90C15, 62J02

UDC: 004.75:519.2

1. Introduction

Permissioned Byzantine Fault Tolerant (BFT) consensus underpins e-governance systems — voting, registries, digital identity — and raises a concrete procurement question: *how many validators should the network have?*

The validator count n pulls in opposite directions. Throughput argues for small n : every validator executes every transaction, and PBFT-family protocols exchange $O(n^2)$ messages per block. Detection reliability argues for large n . This paper develops a reproducible engineering methodology that combines the two under resource-normalized calibration.

A practical obstacle is that dedicated multi-host testbeds are often unavailable at design time. Commodity CI runners are accessible and reproducible, yet pack many validators onto few cores. Without normalization, fitted exponents mix consensus cost with host contention and with the load generator's concurrency schedule. Published single-factor fits $\text{TPS}(n) \propto n^\alpha$ then disagree in magnitude and sometimes in sign. Our earlier studies, extrapolated to sixty validators, differ by roughly an order of magnitude — often read as protocol-specific behaviour. Within the investigated setting we show that it can arise from measurement design alone.

Contributions. The central methodological contribution is resource-normalized measurement (RNM): fixed CPU quota q per validator and $\widehat{\text{TPS}} = \text{TPS}/q$, calibrating consensus cost separately from host contention; in the evaluated deployment it stabilizes the exponent at a fixed quota, although the exponent still depends on the quota level (Section 5). Analytically, under the unsaturated bi-factor model with zero-mean noise, $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ along a concurrency path (Proposition 1) — the standard omitted-variable bias [21], used here to interpret measured single-factor exponents and to show why an RNM factorial design ($\lambda = 0$) matters. We also formulate chance-constrained sizing with feasible set $[n_{\min}, n_{\max}]$ (Proposition 2) and calibrate the workflow on public Besu/QBFT commodity CI (Sections 5–6). The throughput–detection trade-off is formulated but not demonstrated: the measured detector is near chance, the detection constraint is unsatisfiable with it, and the window is set by throughput alone (Section 4.2). Coefficients are deployment-specific and intended for recalibration; cross-client replication is out of scope.

2. Related work

BFT protocols. Classical PBFT [1] tolerates $f < n/3$ at $O(n^2)$ messages [2, 3]. Later designs — HotStuff [4], Tendermint [5], Algorand [6], HoneyBadgerBFT [7] — trade latency and randomization; Vukolić [8] argues that no protocol dominates. These lines of work fix the validator set; they do not size it under joint throughput and detection requirements.

Benchmarks. Fabric [9, 10], BLOCKBENCH [11] and BFT-SMaRt [12] report absolute throughput at chosen configurations. They do not separate concurrency from validator count, nor state measurement conditions under which fitted exponents would be expected to transfer.

Scalability characterization and tools. Amdahl [13], Gustafson [14] and Gunther’s USL [15] fold concurrency and system size into one load variable — the confound that RNM is designed to remove. Concentration bounds [16, 18], Kish’s design effect [17], delta-method intervals [20], the residual bootstrap [19], omitted-variable bias in misspecified regressions [21] and chance-constrained programming [22, 23] are standard instruments.

Prior work. Remote-voting analyses [24, 25, 26] place endpoint and coercion threats out of scope. Our earlier calibration [28] and switching study [29] left open an identifiable consensus exponent and sizing with measured detector rates.

3. Resource-normalized measurement and the bi-factor law

We introduce an isolation protocol, a bi-factor characterization, and an identifiability result under stated assumptions.

3.1. Resource-normalized measurement as an isolation tool

If n validators share a host of Q cores, each receives Q/n cores, so measured throughput mixes algorithmic cost with a shrinking compute budget. RNM is designed to isolate those effects under controlled quotas.

Definition 1 (Resource-normalized measurement). *A campaign follows the RNM protocol with quota q if every validator container receives the same CPU quota and memory limit independently of n , with aggregate reservation nq not exceeding host capacity. Normalized throughput is $\widehat{\text{TPS}} := \text{TPS}/q$.*

Assumption 1 (Separable compute budget). On the operating range of quotas in the campaign, $\text{TPS}(n, c, q) = q \cdot \Phi(n, c) \cdot (1 + r)$ with $|r| \leq Cq$ for a constant C independent of (n, c) there.

Assumption check. Under Assumption 1 and a power-law factor $\Phi \propto n^{-\beta}$, the OLS slope of $\log \text{TPS}$ on $\log n$ at fixed concurrency is independent of q : the relative remainder is absorbed by the intercept. Section 5 tests this invariance across $q \in \{0.20, 0.25, 0.33\}$ and rejects it at $q = 0.33$: RNM stabilizes β at a fixed quota rather than making it quota-free. The calibrated model is reported at $q = 0.25$; outside that regime the coefficients must be re-fit.

Definition 2 (Bi-factor scaling characterization). Under the RNM protocol,

$$\text{TPS}(n, c) = T_0 \cdot g(c) \cdot n^{-\beta}, \quad g(c) = \frac{c^\gamma}{1 + (c/c^*)^\theta}, \quad (1)$$

with $T_0 > 0$, $\beta > 0$, $\gamma \in (0, 1]$, $\theta > \gamma$, $c^* > 0$. In the unsaturated regime $c \ll c^*$ this is log-linear: $\log \text{TPS} = \log T_0 + \gamma \log c - \beta \log n$.

The form separates concurrency from validator-count scaling, reduces to a power law for $c \ll c^*$, saturates at large c , and contains the classical single-factor fit as the case $\gamma = 0$ (or $\lambda = 0$ with $\alpha = -\beta$).

3.2. Identifiability

Proposition 1 is the textbook omitted-variable bias of regressing $\log \text{TPS}$ on $\log n$ while omitting $\log c$ [21], valid under the unsaturated bi-factor model and the stated path and noise assumptions. Its value is interpretive: it shows how to read a measured single-factor exponent under a stated concurrency path. Section 5 checks consistency within the evaluated deployment only.

Proposition 1 (Identifiability of the consensus exponent). Let observations follow the unsaturated bi-factor law with additive noise of mean zero and finite variance, along a concurrency path $\log c_i = \lambda \log n_i + \mu + \eta_i$ ($\mathbb{E}[\eta_i | n_i] = 0$). If the empirical second moment of $x_i = \log n_i$ converges to a positive limit, the OLS slope $\hat{\alpha}_m$ of $\log \text{TPS}_i$ on x_i alone satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$. In particular $\hat{\alpha} > 0 \iff \lambda > \beta/\gamma$, and $\hat{\alpha} = -\beta \iff \lambda = 0$.

Proof. With $x_i = \log n_i$, the path reduces the unsaturated law to $\log \text{TPS}_i = a + \alpha x_i + \varepsilon_i$, $\mathbb{E}[\varepsilon_i | x_i] = 0$, finite variance, and $\alpha = \underbrace{-\beta}_{\text{protocol}} + \underbrace{\gamma\lambda}_{\text{measurement}}$. OLS consistency under the second-moment

condition gives $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \alpha$ [21]; signs follow from $\gamma > 0$. ■

Under the stated model, only a factorial design ($\lambda = 0$) makes the single-factor slope a consistent estimate of $-\beta$. For illustration, hypothetical values $\gamma = 0.8$, $\beta = 0.5$ and $\lambda = 1$ (not the fitted ones) yield $\hat{\alpha} = +0.3$, and extrapolating from $n = 5$ to $n = 60$ differs by a factor ≈ 7 , matching the order of the discrepancy in [28].

Corollary 1 (Non-transferability of single-factor exponents). Under the unsaturated bi-factor model, laboratories measuring along $\lambda_1 \neq \lambda_2$ obtain exponents differing by $\gamma(\lambda_1 - \lambda_2)$. Without reporting λ , published single-factor exponents should not be treated as transferable protocol properties across measurement designs.

Implementation. On a factorial grid ($\lambda = 0$) we estimate $(\log T_0, \gamma, \beta)$ by OLS in log-space and (c^*, θ) by nonlinear least squares on the full c -sweep (Section 5).

4. Prediction intervals, detection, and the sizing problem

The workflow combines a throughput chance constraint with an optional detection constraint; any detector yielding measured (p_d, p_f, ρ) plugs in.

4.1. Definitions and assumptions

Definition 3 (Fault class and detector). A subset B of size k is *faulty* if each member fails a proposal round independently with probability p_d , while honest validators fail with probability $p_f < p_d$ (omission class; equivocation excluded). The detector flags validator i when its failure rate over w rounds exceeds $\tau \in (p_f, p_d)$; detection succeeds when at least one member of B is flagged.

Assumption 2 (Exchangeable correlation). Participation indicators in B are exchangeable with pairwise correlation $\rho \in [0, 1)$, so the mean carries design effect $1 + (k - 1)\rho$ in the sense of Kish [17].

4.2. Prediction intervals and detection bound

With regressor $\mathbf{z} = (1, \log c, -\log n)^\top$ and $\hat{\vartheta} = (\log T_0, \gamma, \beta)^\top$, the delta method [20] gives predictive variance $\hat{V} = \mathbf{z}^\top \widehat{\text{Cov}}(\hat{\vartheta}) \mathbf{z} + \hat{\sigma}^2$. An approximate $(1 - \varepsilon_p)$ prediction interval for $\text{TPS}(n, c)$ is then $\exp\{\log \widehat{\text{TPS}} \mp z_{1-\varepsilon_p/2} \hat{V}^{1/2}\}$; the one-sided lower endpoint $\underline{\text{TPS}}_{1-\varepsilon_p}$ uses $z_{1-\varepsilon_p}$. A residual bootstrap [19] supplies empirical quantiles when residual normality is doubtful. Coverage is checked on held-out validator counts within the calibrated range (Section 5).

With $k = \lceil \varphi n \rceil$, a design-effect heuristic yields the approximate miss probability

$$p_{\text{miss}}(n, w) \lesssim \exp\left\{-\frac{2wk(p_d - \tau)^2}{1 + (k - 1)\rho}\right\}, \quad (2)$$

which is nonincreasing in n and w . The cut $p_{\text{miss}} \leq \varepsilon_s$ gives

$$n_{\min} = \left\lceil \frac{1 + (k - 1)\rho}{\varphi} \cdot \frac{\log(1/\varepsilon_s)}{2w(p_d - \tau)^2} \right\rceil. \quad (3)$$

Expression (2) is an engineering approximation adapting Hoeffding [16] via Kish [17] ($n_{\text{eff}} = kw/(1 + (k - 1)\rho)$), not a new exact bound.

Measured detector. X5 reports per-validator flag rates at the Youden-optimal participation threshold 0.90: $\hat{p}_d = 0.397$ for faulty and $\hat{p}_f = 0.376$ for honest validators ($J = 0.021$). Plugging them into (3) with τ at their midpoint, $\rho = 0$, $\varphi = 0.2$, $w = 30$ and $\varepsilon_s = 0.01$ gives $n_{\min} \approx 3329$, far above any throughput-feasible n . With this detector the detection constraint is unsatisfiable, so the formal feasible set is empty; we relax it (do not enforce it): reported windows use $n_{\min} = 4$ ($3f + 1$ with $f = 1$) and are set by throughput alone. The two-constraint program applies to detectors with J bounded away from zero; its trade-off is not demonstrated here.

4.3. The sizing program

Problem 1 (Validator-set sizing). Given peak demand D_{peak} , levels $\varepsilon_p, \varepsilon_s$, window w , faulty fraction φ , production scale-up $\kappa > 0$ from RNM units to absolute throughput, and

nondecreasing $\text{Cost}(n)$, minimize $\text{Cost}(n)$ subject to $\mathbb{P}[\kappa \cdot \widetilde{\text{TPS}}(n, c^*) \geq D_{\text{peak}}] \geq 1 - \varepsilon_p$ and $p_{\text{miss}}(n, w) \leq \varepsilon_s$, where $c^* = \arg \max g$. When κ is free, take the smallest value that meets the throughput constraint at $n = 4$.

The chance constraint is equivalent, via the prediction interval above, to

$$h(n) := \log(\kappa T_0 g(c^*)/D_{\text{peak}}) - \beta \log n - z_{1-\varepsilon_p} \hat{V}^{1/2}(n, c^*) \geq 0. \quad (4)$$

On the model-based extrapolation range ($n \geq 16$) the fitted covariance makes $\hat{V}$ nondecreasing in $\log n$, so with $\beta > 0$ the map h is strictly decreasing there; on the measured grid we evaluate h directly.

Proposition 2 (Feasibility window). *Whenever h is strictly decreasing and the approximate detection cut (3) is used, the feasible set is $\{n : n_{\min} \leq n \leq n_{\max}\}$ with $n_{\min}$ from (3) and $n_{\max} = \lfloor \sup\{n : h(n) \geq 0\} \rfloor$. If nonempty and Cost is nondecreasing then $n^* = n_{\min}$.*

Proof. Detection holds on $\{n \geq n_{\min}\}$ and throughput on $\{n \leq n_{\max}\}$, hence an interval; monotone cost selects the left endpoint. ■

Corollary 2 (Empty feasible set). *If $n_{\min} > n_{\max}$, no validator count satisfies both requirements under the stated constraints: the architecture must change, not merely the provisioning.*

Protocol-efficient sizes are $n = 3f + 1$; other counts appear on the measurement grid, but a procurement decision intersects the window with $\{3f + 1\}$. The window is the intersection of two monotone constraints [22, 23]. Within the proposed methodology it converts a calibrated model into a reproducible sizing workflow for the evaluated deployment class.

5. Experimental campaign on commodity CI

5.1. Setup and design

The campaign runs in a public continuous-integration workflow on a four-core, 16 GB hosted runner. Constrained CI is the intended calibration setting: accessible, reproducible, and usable when dedicated clusters are unavailable. Each job provisions a Hyperledger Besu [27] QBFT network of n validators under the RNM protocol (Definition 1). A closed-loop generator drives c concurrent submitters and records steady-state committed throughput after warm-up. Faults are injected with kernel traffic control (omission and duplication; not equivocation).

X1/X3 calibrate (β, γ) on a factorial grid; **X2** checks the quota regime; **X3b** re-samples along concurrency paths; **X4** varies emulated RTT; **X5** reports detector rates; **X7/X8** check prediction-interval coverage and sizing ablation. Cross-client replication (**X9/GoQuorum**) is out of scope. In total 94 successful records, ≈ 27.2 core-hours, $n_{\max} = 16$.

Reproducibility. All experiments are reproducible from the public CI workflow and archived artefacts at https://github.com/bpenyak/Resource_Normalized_Scaling_Laws_for_Byzantine_Consensus.

5.2. Results

X1/X3 – bi-factor calibration. Joint OLS on the 48 unsaturated fault-free runs at the reference quota ($n \in \{4, \dots, 16\}$, $c \in \{1, \dots, 16\}$) yields $\beta = 1.651 \pm 0.151$ and $\gamma = 0.178 \pm 0.070$

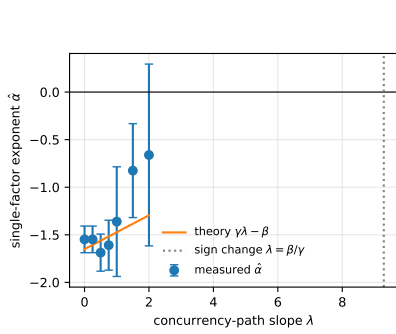


Fig. 1. Measured $\hat{\alpha}$ (points) vs. analytical $\gamma\lambda - \beta$ (line) (Proposition 1); predicted sign change at $\lambda = 9.29$, outside the tested paths

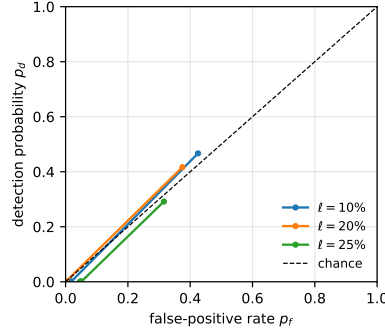


Fig. 2. Measured detector ROC (AUC = 0.506, near chance); the detection constraint is unsatisfiable with it

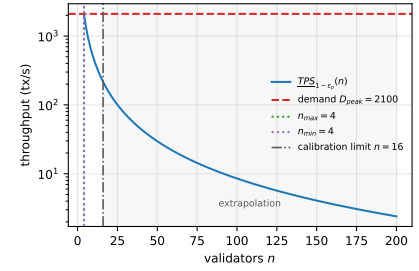


Fig. 3. Scenario A: lower PI throughput vs. n at $\kappa = 13.0$; window $[4, 4]$; dash-dotted: calibration limit $n = 16$

($R^2 = 0.727$; Table 1) at fixed concurrency and $q = 0.25$, using all counts including the hold-out ones. A constant β averages a steepening slope: locally $\approx 1.4, 1.7, 2.8$ over $n = 4-7, 7-13, 13-16$ at $c = 16$, and refits on $n \leq 9$ give 1.300 (pooled quotas) and 1.196 (reference quota), 2.3 and 3.0 standard errors lower. We attribute this to host contention as nq approaches the four host cores shared with the load generator ($nq = 4$ at $n = 16$): RNM removes contention only while nq stays below the host cores. The headline β is conservative for $n_{\max}$ but optimistic for κ (13.0 vs. 16.7 under the $n \leq 9$ refit); the window $[4, 4]$ holds under either fit by construction.

X3b – path check of Proposition 1. A single-factor fit along $\lambda = 1$ gives $\hat{\alpha} = -1.362$ against $\gamma\lambda - \beta = -1.473$. Figure 1 shows measured exponents tracking $\gamma\lambda - \beta$ – consistency with the analytical prediction, not universal validity of the bi-factor form. The predicted sign change ($\lambda \approx 9.29$) lies outside the tested $\lambda \in \{0, 1, 2\}$, so the check tests the size of the shift, not the sign flip.

X2 – quota regime. X2 measures $n \in \{4, 7, 13\}$ at $c = 16$. The partial F -test for quota $\times \log n$ interaction rejects a common slope ($p = 0.002$): β is 1.550 and 1.541 at $q = 0.20, 0.25$ but 0.879 at $q = 0.33$. In the $q = 0.33$ cells, throughput at $n = 4$ no longer grows with the quota (55.5, 68.6, 70.0 TPS), suggesting a non-CPU ceiling that breaks Assumption 1, and at $n = 13$ the reservation of ≈ 4.3 cores exceeds the host, outside Definition 1. RNM thus stabilizes β at a fixed quota in the CPU-bound regime but does not make it quota-free; coefficients are reported at $q = 0.25$.

X4 – latency. On its own grid ($n \in \{7, 13\}$, $c = 16$, $q = 0.25$), X4 gives $\beta = 1.459$ at 0 ms and 2.708 at 200 ms RTT ($\Delta\beta = 1.248$). The baseline is a two-point slope from four runs and agrees with the joint fit $\beta = 1.651$ within 1.3 standard errors; intermediate RTTs are non-monotone (Table 1), so X4 indicates the direction and rough size of the WAN penalty, not a latency law.

X5 – detector. Over 13 fault-injection runs the pooled ROC has AUC = 0.506 (Figure 2, near chance); ρ is not identifiable from these runs and is set to 0. With these rates the detection constraint is unsatisfiable and is relaxed, not enforced (Section 4.2).

X7/X8 – coverage, comparison, and ablation. A separate refit on the 45 fault-free runs with $n \leq 9$, which never sees the hold-out runs, gives intervals covering 20 and 21 of the 22 runs at $n \in \{11, 13, 16\}$ at the 90% and 95% levels. The check is coarse (most runs at $n = 13$, pooled X2 quotas, binomial standard error ≈ 6 points; $n = 11$ is not $3f + 1$) and validates

Table 1. Calibrated parameters, quota check, detector, and coverage (evaluated deployment)

Quantity	Estimate	Note
$T_0; \beta; \gamma$	2016.5; 1.651 ± 0.151 ; 0.178 ± 0.070	bi-factor
$c^*, \theta; \hat{\sigma}$	67.9, 2.542; 0.404	saturation / residual
$\hat{\alpha}$ at $\lambda = 0, 1, 2$	-1.548, -1.362, -0.661	X3b
β at $q = 0.20, 0.25, 0.33$	1.550, 1.541, 0.879	X2, $p = 0.002$
β at RTT 0, 25, 50, 100, 200 ms	1.459, 1.925, 0.252, 2.011, 2.708	X4, $n \in \{7, 13\}$
$\hat{p}_d, \hat{p}_f$ (flag rates); J ; AUC	0.397, 0.376; 0.021; 0.506	X5, threshold 0.90
coverage 90%/95%	20/22, 21/22	fit $n \leq 9$; hold-out $n \in \{11, 13, 16\}$

the interval procedure, not the headline β . Out-of-sample log-RMSE for linear / single-factor / USL [15] / bi-factor predictors is 1.329, 0.523, 0.637, 0.538; single-factor is slightly better here only because all held-out runs share $c = 16$. Relative sizing costs: 4.00, 1.50, 1.50, 1.00.

6. Application, implications, and limitations

6.1. Application and what-if analysis

We instantiate Problem 1 on $\approx 3 \times 10^7$ ballots over twelve hours: a mean of ≈ 694 TPS and, with an assumed peak-to-mean factor of ≈ 3 (not a measured profile), a peak $D_{\text{peak}} = 2100$ TPS. The detection requirement — a coordinated transaction-omission fault of fraction $\varphi = 0.2$ within one epoch — is unsatisfiable and is relaxed, not enforced (Section 4.2), so $n_{\min} = 4$. Fault-free CI runs without emulated latency commit at most ≈ 89 TPS (at $n = 7$, $c = 16$, $q = 0.25$); the model works in RNM units $\widetilde{\text{TPS}} = \text{TPS}/q$ (throughput per core of per-validator quota), and under Assumption 1 the scale-up κ is the per-validator core count of the target hardware class. Linearity of throughput in the quota is the key assumption and the main extrapolation risk: $\kappa = 13.0$ cores is ≈ 52 times the calibrated $q = 0.25$, whereas X2 already shows throughput ceasing to grow at $q = 0.33$. κ is therefore a hardware requirement to verify on the target hardware, not a measured capability. With free κ , Problem 1 takes

$$\kappa^* = \frac{D_{\text{peak}}}{\widetilde{\text{TPS}}_{1-\varepsilon_p}(4, c^*)}, \quad (5)$$

the minimal scale at which the lower prediction-interval bound at $n = 4$ meets D_{peak} .

Scenario A (baseline). Substituting the calibrated coefficients ($\beta = 1.651$, $\gamma = 0.178$) at $c^* = 32$ yields $\kappa = 13.0$ (cores per validator) and feasibility window $[4, 4]$ with $n^* = 4$ (Proposition 2; Figure 3). The single count is by construction: κ is the smallest scale admitting $n = 4$, so $h(4) = 0$ and, with $\beta > 0$, $n_{\max} = 4$; the informative output is κ . (c^* maximizes the fitted g beyond the OLS grid $c \leq 16$; at $c = 16$, $\kappa = 12.94$.)

Scenarios B and C (fixed hardware class). Hold $\kappa = 13.0$. Doubling D_{peak} gives $h(4) = -\log 2 < 0$; raising β by the within-X4 WAN increment $\Delta\beta = 1.248$ lowers $h(4)$ by $\Delta\beta \log 4$. Both windows are empty (Corollary 2), and adding validators cannot help because predicted throughput falls with n . Both follow from the zero margin of Scenario A; their content is the required scale-up: $\kappa = 26.0$ and ≈ 73.4 for doubled demand and WAN, respectively.

The same workflow applies to registry and identity sizing after recalibration; binding elections on a blockchain remain outside the recommendation [24, 25, 26].

6.2. What this implies

A fitted (β, γ) is Besu/QBFT- and deployment-specific; transfer is appropriate only when concurrency path, quota regime, and RTT class match the calibration, while the RNM protocol and the workflow transfer via recalibration. Benchmarks should report λ (or $\lambda = 0$), per-validator quota, and RTT class (Corollary 1).

6.3. Limitations

Fault class and detection. Encrypted transport limits X5 to omission/duplication, and the resulting detector is near chance; the detection side is therefore not exercised empirically, and demonstrating the throughput–detection trade-off needs a detector with J bounded away from zero.

Infrastructure. Four-core CI with $n \leq 16$ is the intended substrate; at $n = 16$ the reservation equals the four host cores. Beyond measured counts, predictions are model-based.

7. Conclusions

Analytical results. Under the unsaturated bi-factor model with zero-mean, finite-variance noise and the stated path assumptions, the single-factor OLS exponent satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ (Proposition 1), an omitted-variable bias. Opposing monotone constraints yield a closed integer feasibility window (Proposition 2).

Experimentally supported observations. Under RNM on four-core commodity CI, $\beta = 1.651 \pm 0.151$ and $\gamma = 0.178 \pm 0.070$ at $q = 0.25$; β is stable at a fixed quota but quota-dependent; path re-sampling agrees with $\gamma\lambda - \beta$. The detector is near chance, so the window is throughput-only. Doubled demand or WAN β requires $\kappa = 26.0$ or ≈ 73.4 , respectively, instead of 13.0; these values rest on the untested linearity of throughput in the quota. Numerical results are deployment-specific.

Engineering implications. We contribute a reproducible methodology for calibration and capacity planning under resource normalization; the throughput–detection trade-off is formulated but, with the measured detector, not demonstrated. Other deployments require recalibration of (β, γ) and (p_d, p_f, ρ) while reusing the same workflow and public artefacts.

Conflict of interest

The authors declare no conflict of interest.

Funding

The research was conducted without financial support.

Data availability

Workflow definition, generators, fault-injection and analysis code, and raw artefacts are available at https://github.com/bpenyak/Resource_Normalized_Scaling_Laws_for_Byzantine_Consensus.

Use of artificial intelligence

AI-assisted tools were used for language editing and typesetting. All scientific results, proofs, design and analysis were produced by the authors.

Contributions of authors

Peniak B. O.: conceptualization, methodology, formal analysis, software, investigation, data curation, writing — original draft, visualization. **Liubinskiy B. B.:** supervision, validation, writing — review and editing.

-
- [1] Castro M., Liskov B. Practical Byzantine fault tolerance. Proceedings of the Third Symposium on Operating Systems Design and Implementation (OSDI). 173–186 (1999). <https://www.usenix.org/conference/osdi-99/practical-byzantine-fault-tolerance>
 - [2] Lamport L., Shostak R., Pease M. The Byzantine generals problem. ACM Transactions on Programming Languages and Systems. **4** (3), 382–401 (1982). <https://doi.org/10.1145/357172.357176>
 - [3] Dwork C., Lynch N., Stockmeyer L. Consensus in the presence of partial synchrony. Journal of the ACM. **35** (2), 288–323 (1988). <https://doi.org/10.1145/42282.42283>
 - [4] Yin M., Malkhi D., Reiter M. K., Gueta G. G., Abraham I. HotStuff: BFT consensus with linearity and responsiveness. Proceedings of the ACM Symposium on Principles of Distributed Computing (PODC). 347–356 (2019). <https://doi.org/10.1145/3293611.3331591> <https://arxiv.org/abs/1803.05069>
 - [5] Buchman E. Tendermint: Byzantine fault tolerance in the age of blockchains. Master’s thesis, University of Guelph (2016). <https://hdl.handle.net/10214/9769>
 - [6] Gilad Y., Hemo R., Micali S., Vlachos G., Zeldovich N. Algorand: scaling Byzantine agreements for cryptocurrencies. Proceedings of the 26th Symposium on Operating Systems Principles (SOSP). 51–68 (2017). <https://doi.org/10.1145/3132747.3132757> <https://arxiv.org/abs/1607.01341>
 - [7] Miller A., Xia Y., Croman K., Shi E., Song D. The honey badger of BFT protocols. Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS). 31–42 (2016). <https://doi.org/10.1145/2976749.2978399> <https://eprint.iacr.org/2016/199>
 - [8] Vukolić M. The quest for scalable blockchain fabric: proof-of-work vs. BFT replication. Open Problems in Network Security. IFIP AICT, vol. 468. Springer. 112–125 (2015). https://doi.org/10.1007/978-3-319-39028-4_9
 - [9] Androulaki E., et al. Hyperledger Fabric: a distributed operating system for permissioned blockchains. Proceedings of the Thirteenth EuroSys Conference. 30:1–30:15 (2018). <https://doi.org/10.1145/3190508.3190538> <https://arxiv.org/abs/1801.10228>
 - [10] Thakkar P., Nathan S., Viswanathan B. Performance benchmarking and optimizing Hyperledger Fabric blockchain platform. 26th IEEE International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS). 264–276 (2018). <https://doi.org/10.1109/MASCOTS.2018.00034>
 - [11] Dinh T. T. A., Wang J., Chen G., Liu R., Ooi B. C., Tan K.-L. BLOCKBENCH: a framework for analyzing private blockchains. Proceedings of the ACM International Conference on Management of Data (SIGMOD). 1085–1100 (2017). <https://doi.org/10.1145/3035918.3064033> <https://arxiv.org/abs/1703.04057>

- [12] Sousa J., Bessani A., Vukolić M. A Byzantine fault-tolerant ordering service for the Hyperledger Fabric blockchain platform. 48th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN). 51–58 (2018). <https://doi.org/10.1109/DSN.2018.00018>
- [13] Amdahl G. M. Validity of the single processor approach to achieving large scale computing capabilities. Proceedings of the AFIPS Spring Joint Computer Conference. 483–485 (1967). <https://doi.org/10.1145/1465482.1465560>
- [14] Gustafson J. L. Reevaluating Amdahl’s law. Communications of the ACM. **31** (5), 532–533 (1988). <https://doi.org/10.1145/42411.42415>
- [15] Gunther N. J. Guerrilla capacity planning: a tactical approach to planning for highly scalable applications and services. Springer (2007). <https://doi.org/10.1007/978-3-540-31010-5>
- [16] Hoeffding W. Probability inequalities for sums of bounded random variables. Journal of the American Statistical Association. **58** (301), 13–30 (1963). <https://doi.org/10.1080/01621459.1963.10500830>
- [17] Kish L. Survey sampling. John Wiley & Sons, New York (1965). <https://archive.org/details/surveysampling00kish>
- [18] Boucheron S., Lugosi G., Massart P. Concentration inequalities: a nonasymptotic theory of independence. Oxford University Press (2013). <https://doi.org/10.1093/acprof:oso/9780199535255.001.0001>
- [19] Efron B., Tibshirani R. J. An introduction to the bootstrap. Chapman and Hall (1993). <https://doi.org/10.1201/9780429246593>
- [20] Seber G. A. F., Wild C. J. Nonlinear regression. Wiley (2003). <https://doi.org/10.1002/0471725315>
- [21] Wooldridge J. M. Econometric analysis of cross section and panel data. 2nd ed. MIT Press (2010). <https://mitpress.mit.edu/9780262232586/>
- [22] Charnes A., Cooper W. W. Chance-constrained programming. Management Science. **6** (1), 73–79 (1959). <https://doi.org/10.1287/mnsc.6.1.73>
- [23] Nemirovski A., Shapiro A. Convex approximations of chance constrained programs. SIAM Journal on Optimization. **17** (4), 969–996 (2006/2007). <https://doi.org/10.1137/050622328> <https://optimization-online.org/wp-content/uploads/2004/12/1033.pdf>
- [24] Springall D., Finkenauer T., Durumeric Z., Kitcat J., Hursti H., MacAlpine M., Halderman J. A. Security analysis of the Estonian internet voting system. Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS). 703–715 (2014). <https://doi.org/10.1145/2660267.2660315> <https://jhalderm.com/pub/papers/ivoting-ccs14.pdf>
- [25] Specter M. A., Koppel J., Weitzner D. The ballot is busted before the blockchain: a security analysis of Voatz. Proceedings of the 29th USENIX Security Symposium. 1535–1553 (2020). <https://www.usenix.org/system/files/sec20-specter.pdf>
- [26] Park S., Specter M., Narula N., Rivest R. L. Going from bad to worse: from internet voting to blockchain voting. Journal of Cybersecurity. **7** (1), tyaa025 (2021). <https://doi.org/10.1093/cybsec/tyaa025>
- [27] Hyperledger Besu documentation. Hyperledger Foundation. <https://github.com/hyperledger/besu> <https://www.hyperledger.org/use/besu>
- [28] Peniak B. O., Liubinskiy B. B., Solomka I. R. Minimal-sample performance prediction for Byzantine consensus: the α -calibration method. Mathematical Modeling and Computing. (accepted; to appear) (2026). <https://science.lpnu.ua/mmc>
- [29] Peniak B. O., Liubinskiy B. B. Adaptive hybrid consensus mechanism for blockchain-based e-governance: a machine learning approach. Scientific Bulletin of Uzhhorod University. Series of

Mathematics and Informatics. **49** (2), 255–261 (2026). [https://doi.org/10.24144/2616-7700.2026.49\(2\).255-261](https://doi.org/10.24144/2616-7700.2026.49(2).255-261)

РЕСУРСНО-НОРМОВАНІ ЗАКОНИ МАСШТАБУВАННЯ BFT-КОНСЕНСУСУ: ІМОВІРНІСНО-ОБМЕЖЕНЕ ВИЗНАЧЕННЯ РОЗМІРУ МНОЖИНИ ВАЛІДАТОРІВ ЗА ВИМІРЮВАННЯМИ НА ТИПОВІЙ СІ-ІНФРАСТРУКТУРІ

Пеняк Б. О., Любінський Б. Б.

*Національний університет «Львівська політехніка»
вул. С. Бандери, 12, 79013, Львів, Україна*

Визначення розміру BFT-множини валідаторів для електронного урядування узгоджує пропускну здатність, що спадає з n , із силою виявлення відмов, що з n зростає. Запропоновано ресурсно-нормовані вимірювання з фіксованою CPU-квотою валідатора; показано, що за ненасиченої двофакторної моделі з шумом нульового середнього однофакторний показник МНК прямує до $\gamma\lambda - \beta$ (зміщення через пропущену змінну); сформульовано задачу з імовірнісними обмеженнями та вікном $[n_{\min}, n_{\max}]$. На чотириядерній публічній СІ ($n \leq 16$) показник стабільний за фіксованої квоти, але залежить від її рівня; вимірний детектор близький до випадкового ($AUC = 0.506$), тож детекційне обмеження послаблено (не застосовується), і вікно визначає лише пропускну здатність. Коефіцієнти специфічні для розгортання.

Ключові слова: візантійська відмовостійкість, консенсус, закон масштабування, планування потужності, оптимізація з імовірнісними обмеженнями, електронне урядування

УДК: 004.75:519.2